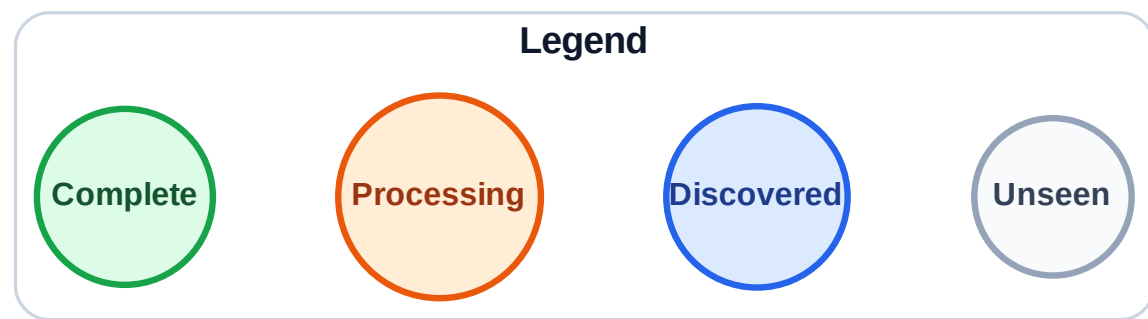
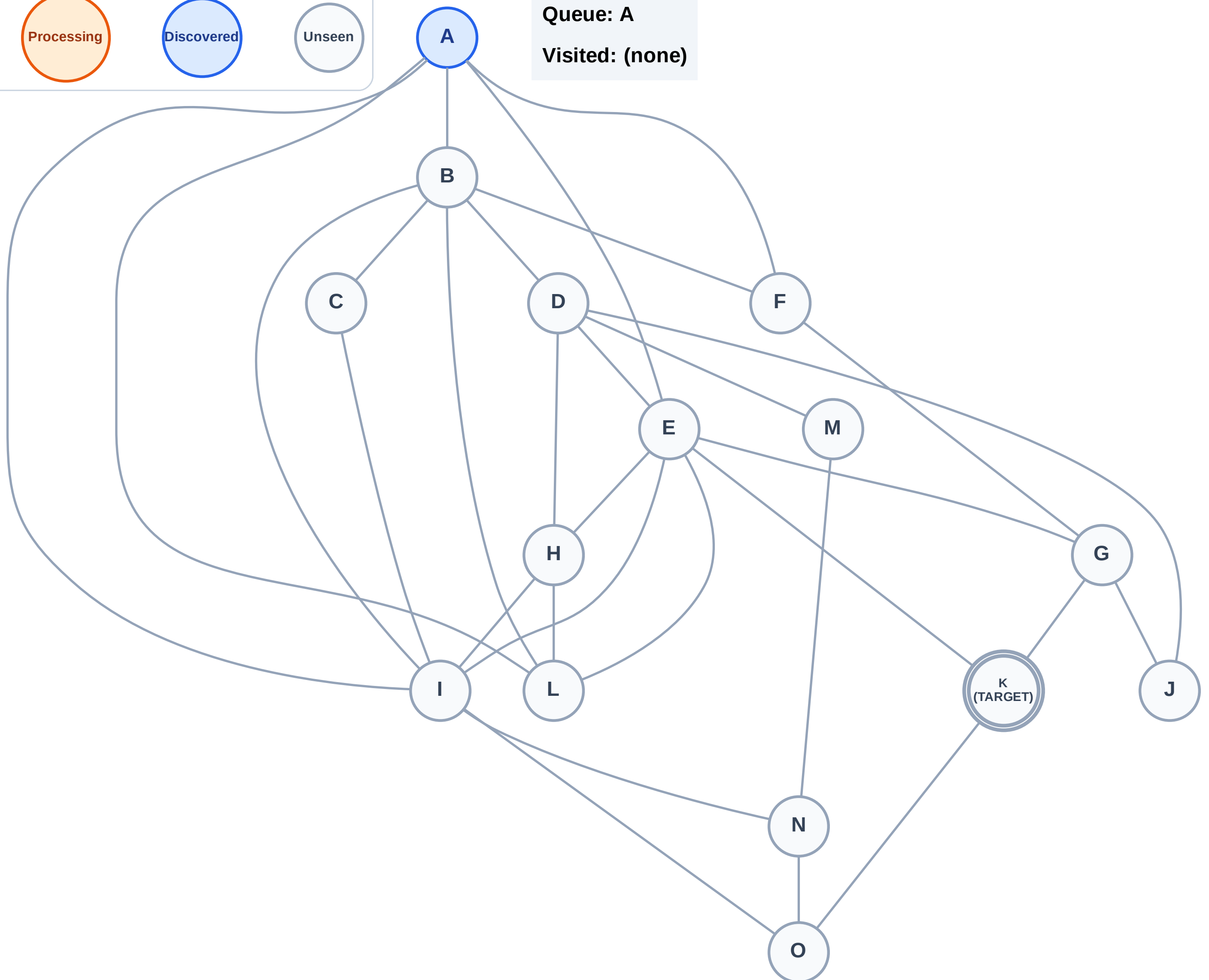


BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

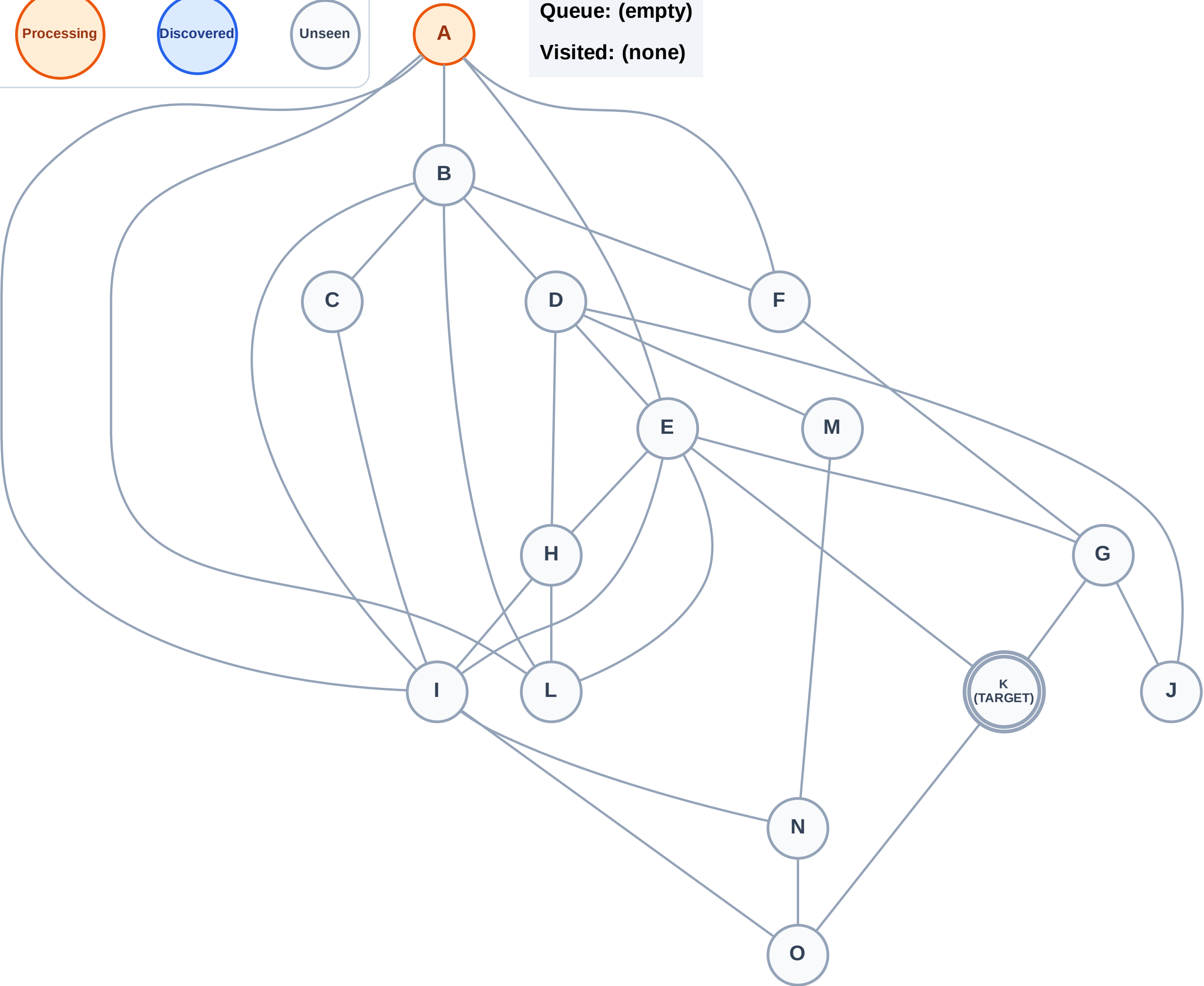
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



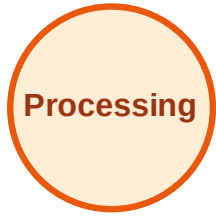
BFS Traversal

Step 3: Discover B, E, F, I, L from A

Legend



Complete



Processing



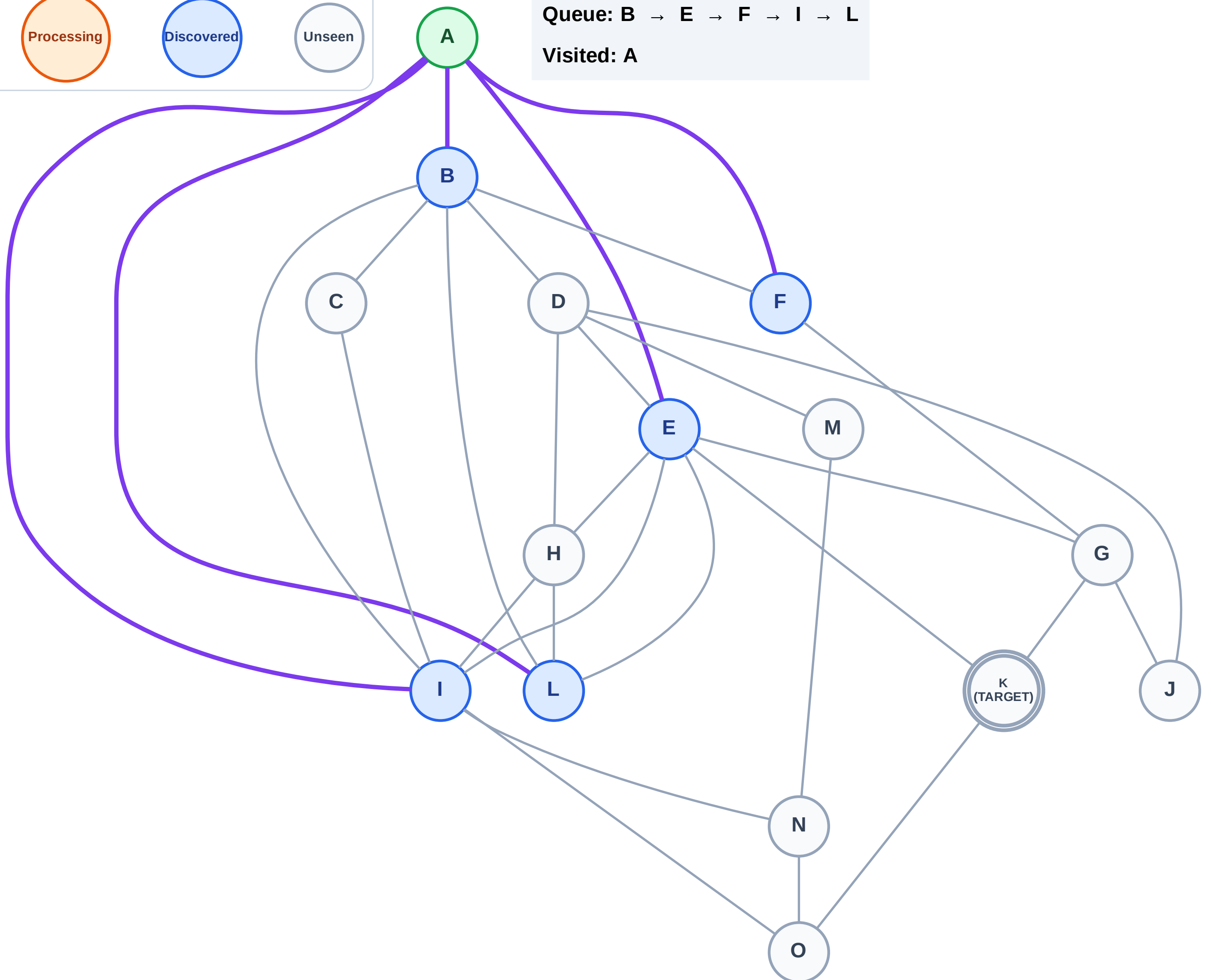
Discovered



Unseen

Queue: B → E → F → I → L

Visited: A



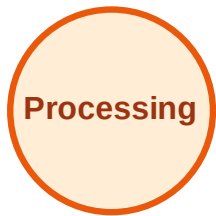
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



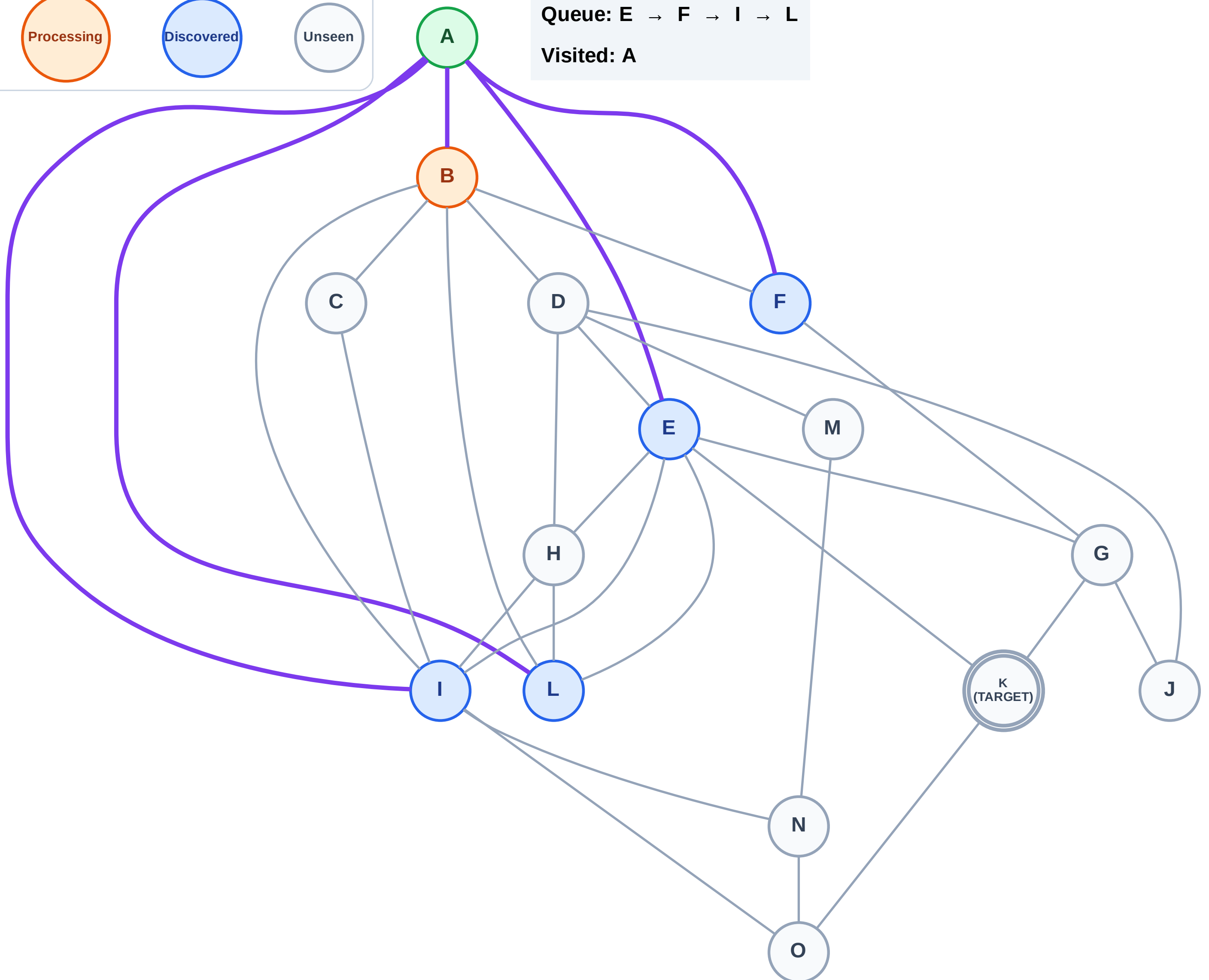
Discovered



Unseen

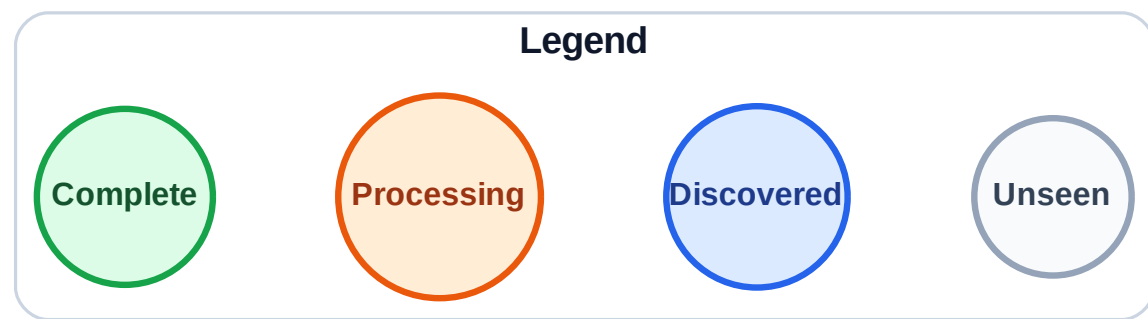
Queue: E → F → I → L

Visited: A

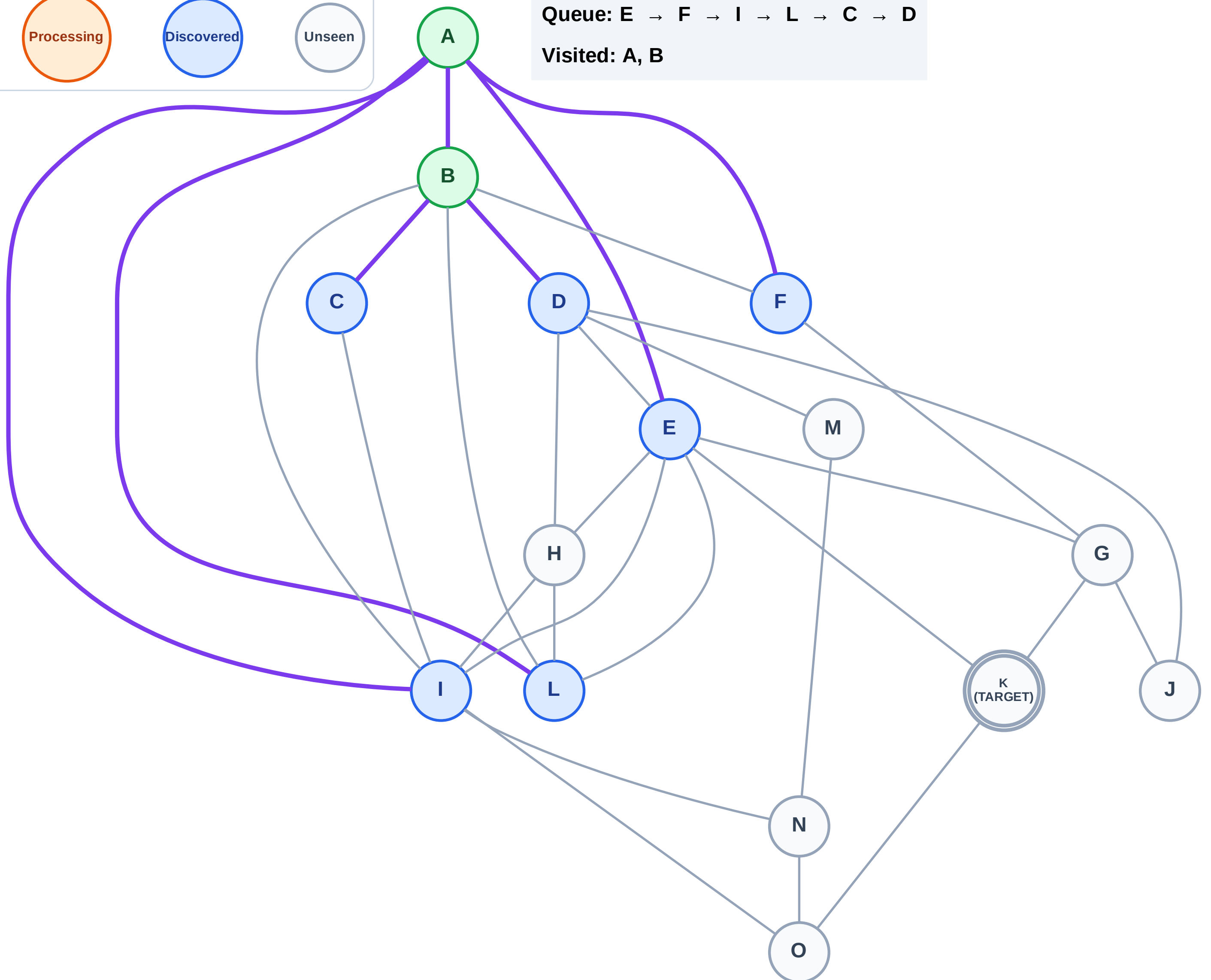


BFS Traversal

Step 5: Discover C, D from B



Queue: E → F → I → L → C → D
Visited: A, B



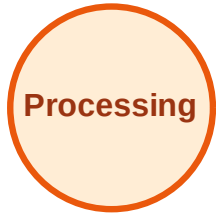
BFS Traversal

Step 6: Process node E

Legend



Complete



Processing



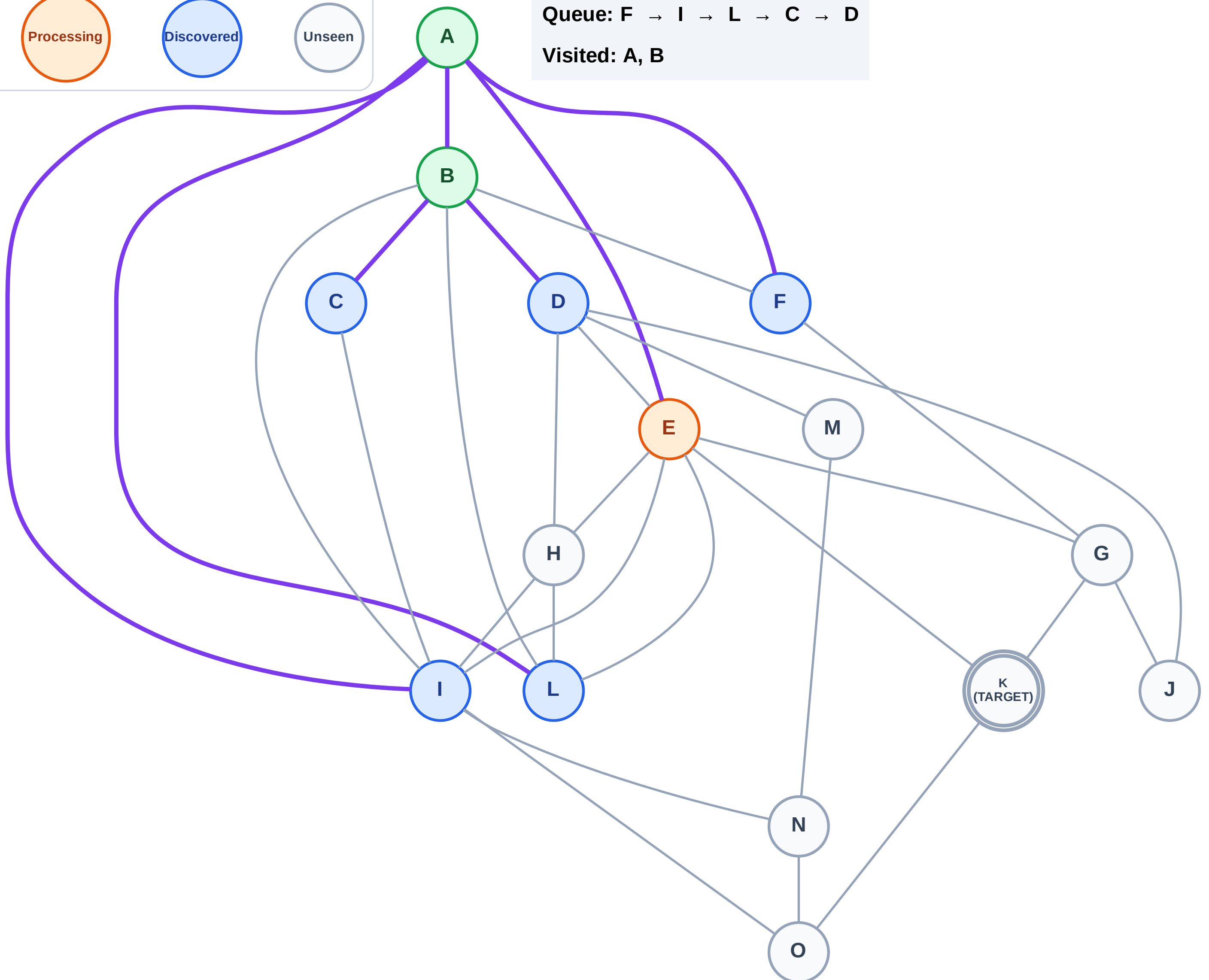
Discovered



Unseen

Queue: F → I → L → C → D

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → E → K

Legend

Complete

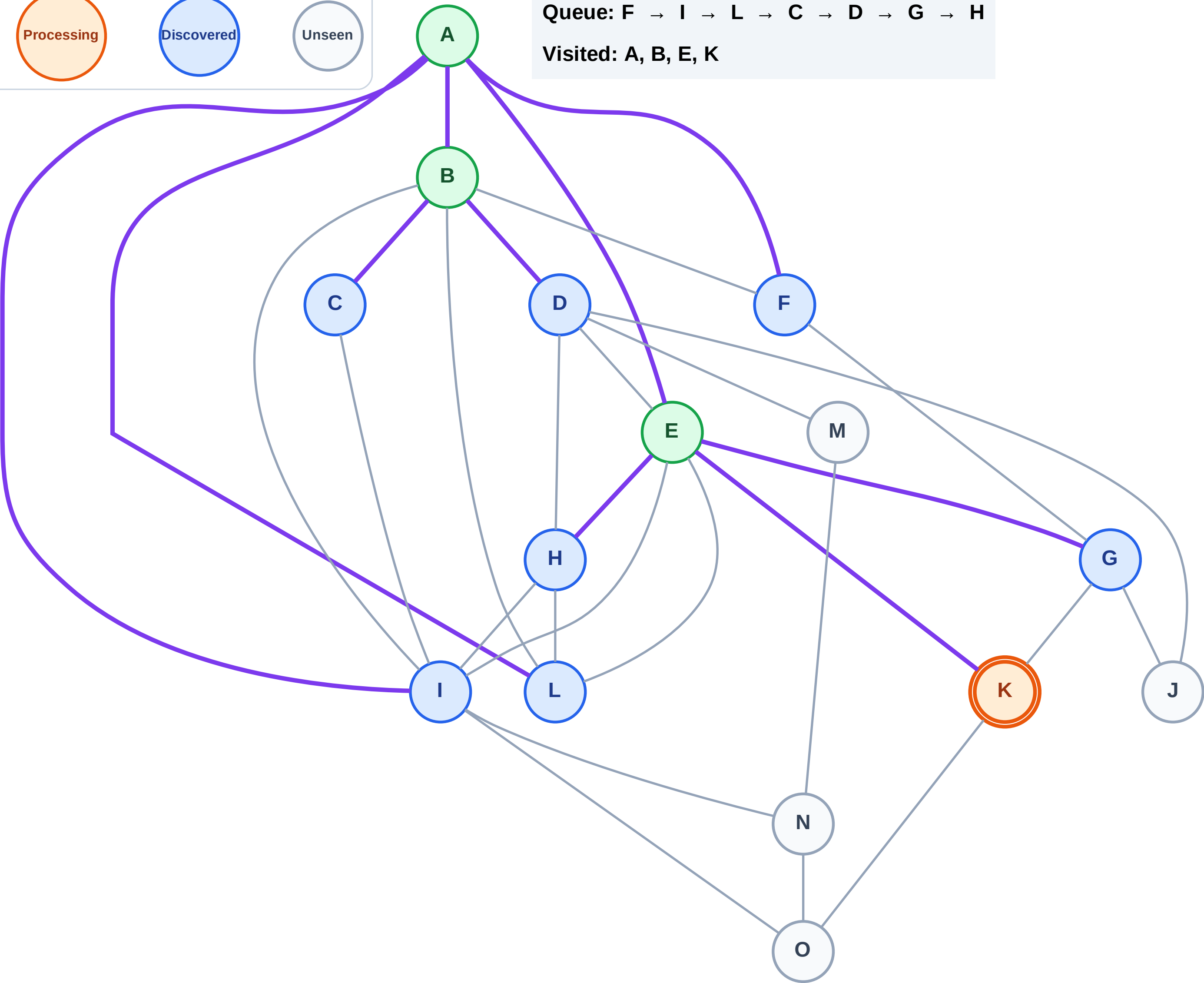
Processing

Discovered

Unseen

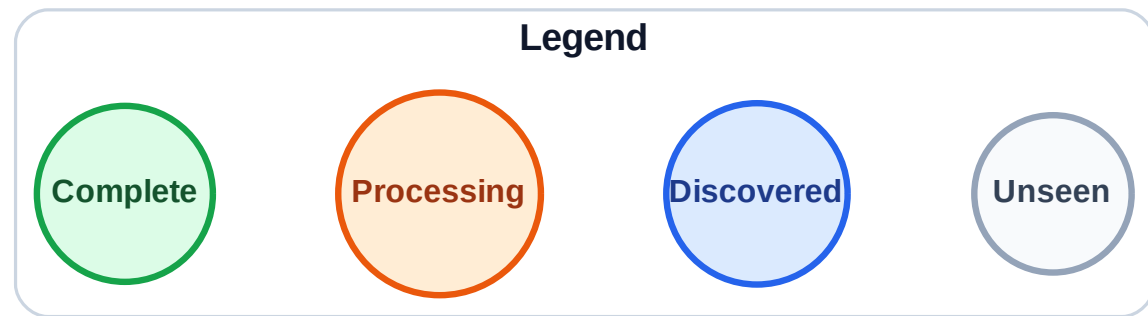
Queue: F → I → L → C → D → G → H

Visited: A, B, E, K

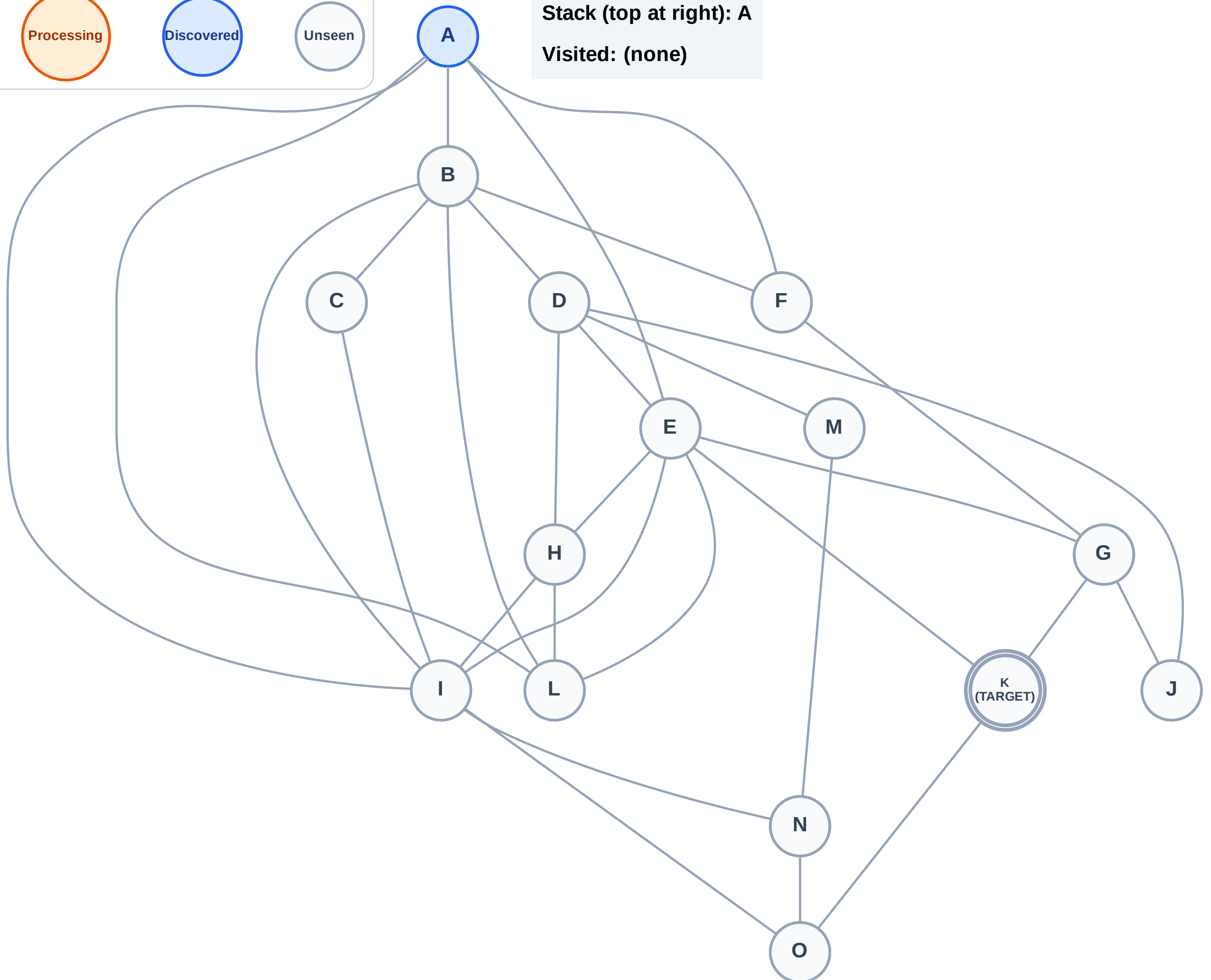


DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

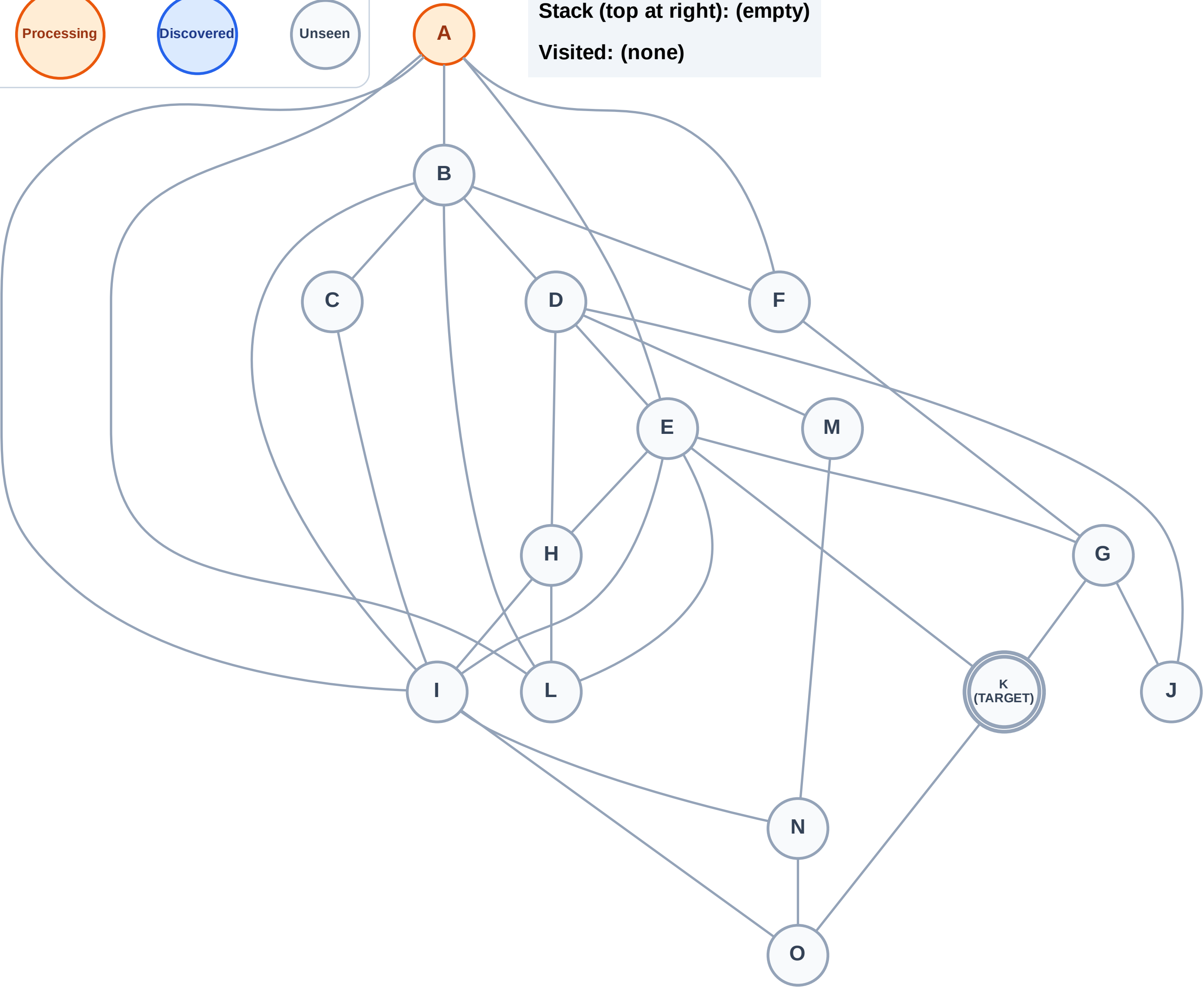
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



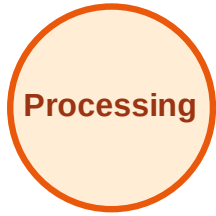
DFS Traversal

Step 3: Discover L, I, F, E, B from A

Legend



Complete



Processing



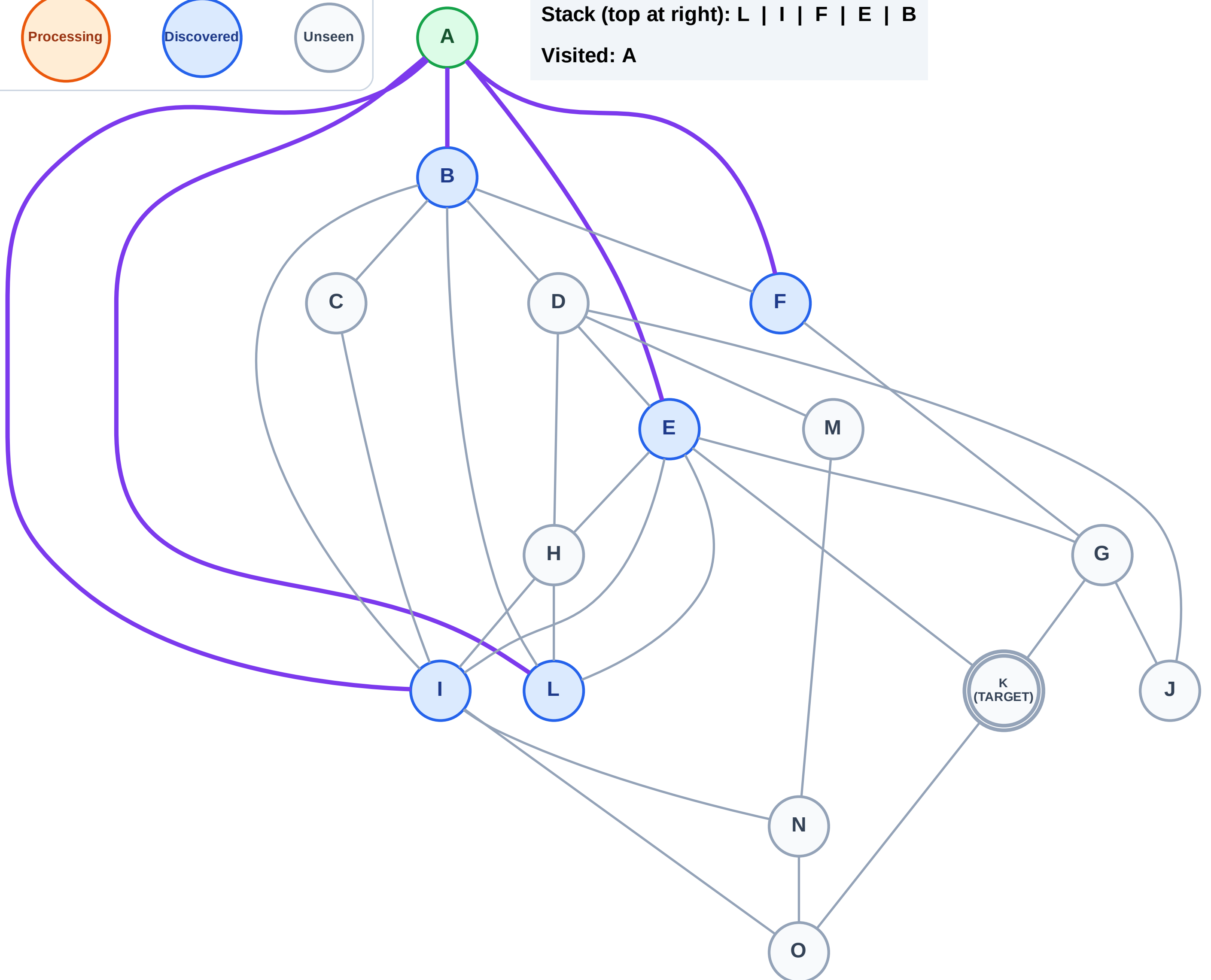
Discovered



Unseen

Stack (top at right): L | I | F | E | B

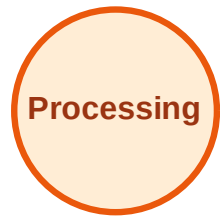
Visited: A



DFS Traversal

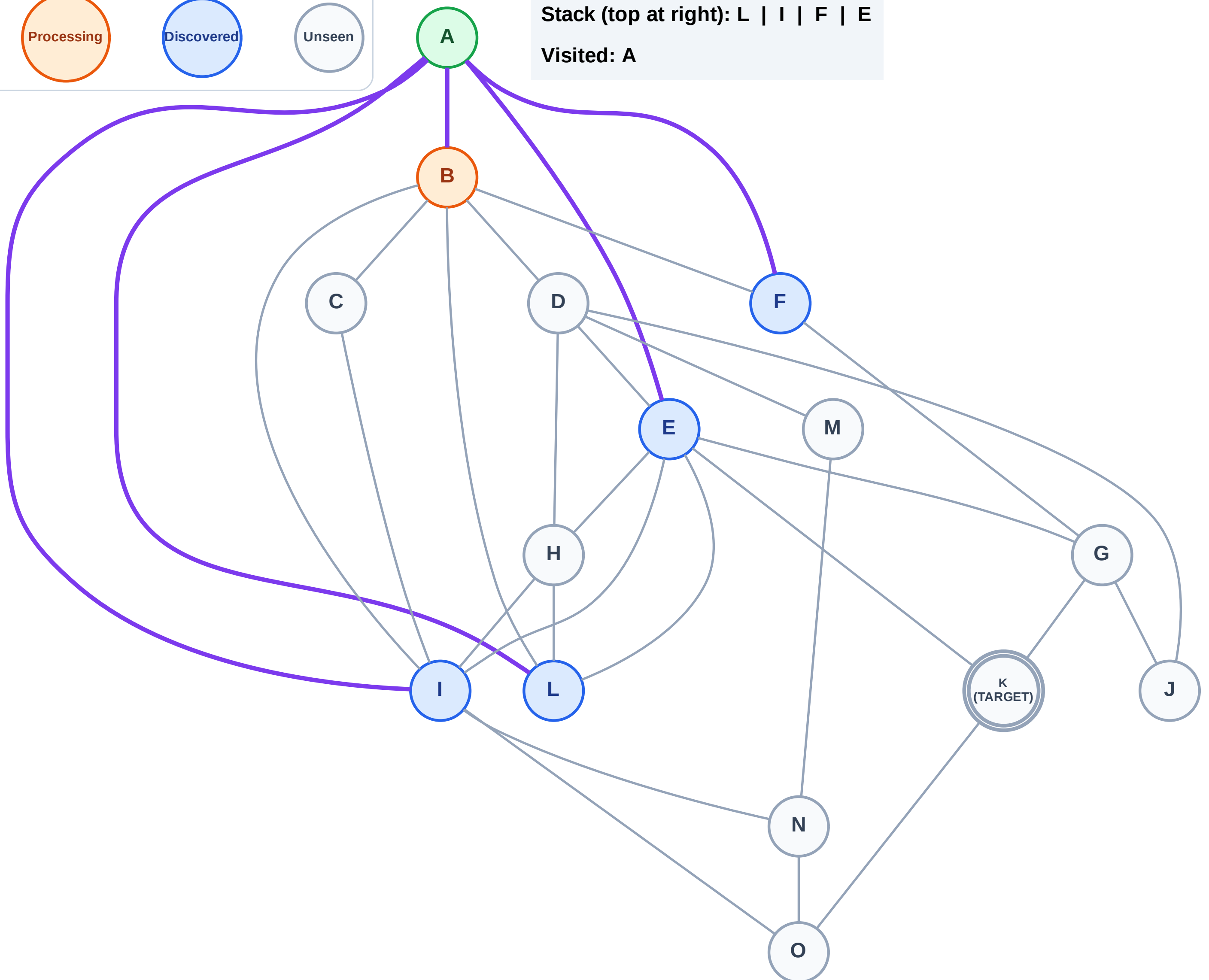
Step 4: Process node B

Legend



Stack (top at right): L | I | F | E

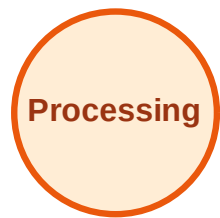
Visited: A



DFS Traversal

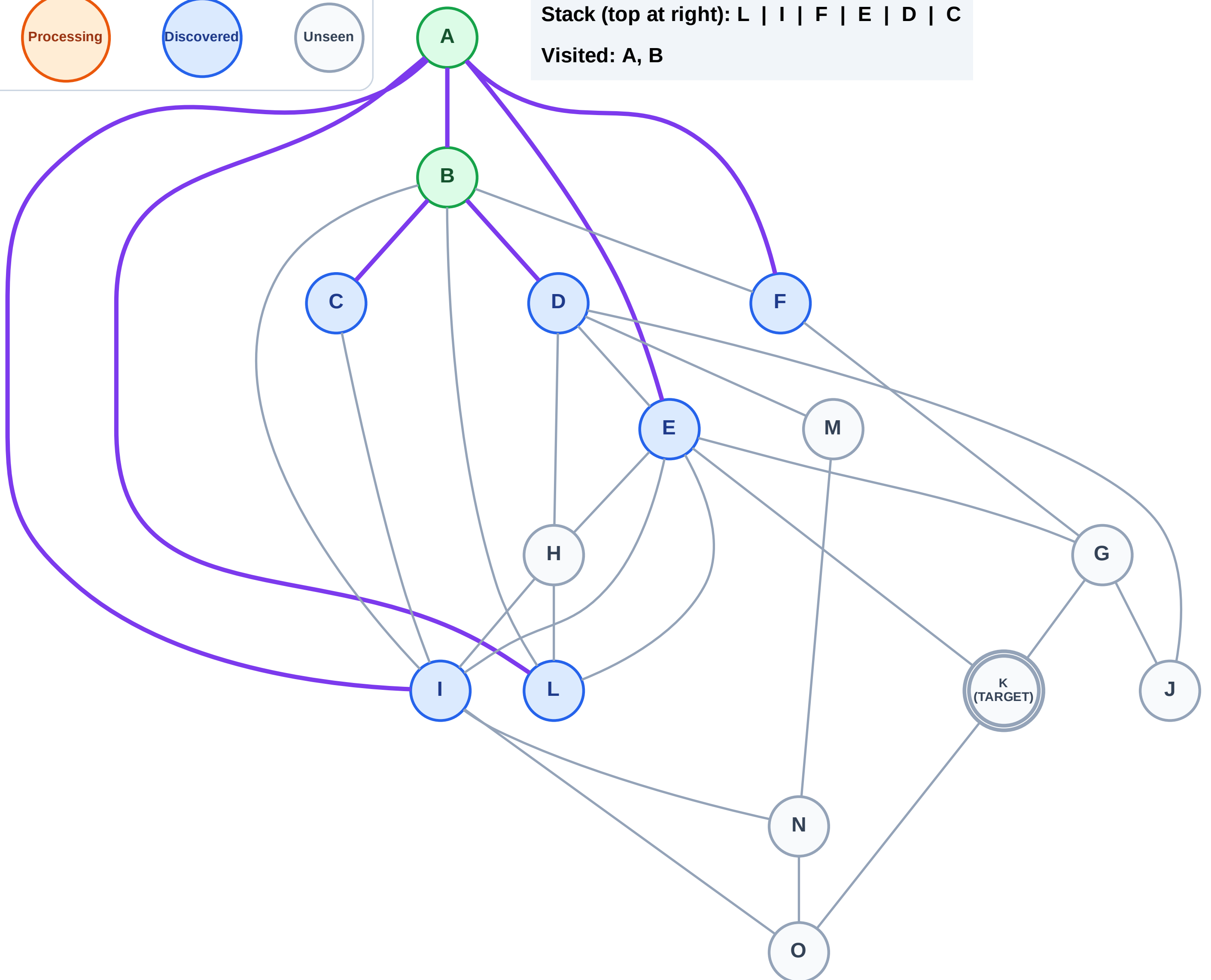
Step 5: Discover D, C from B

Legend



Stack (top at right): L | I | F | E | D | C

Visited: A, B



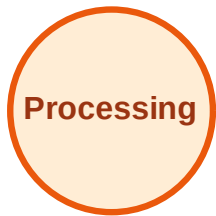
DFS Traversal

Step 6: Process node C

Legend



Complete



Processing



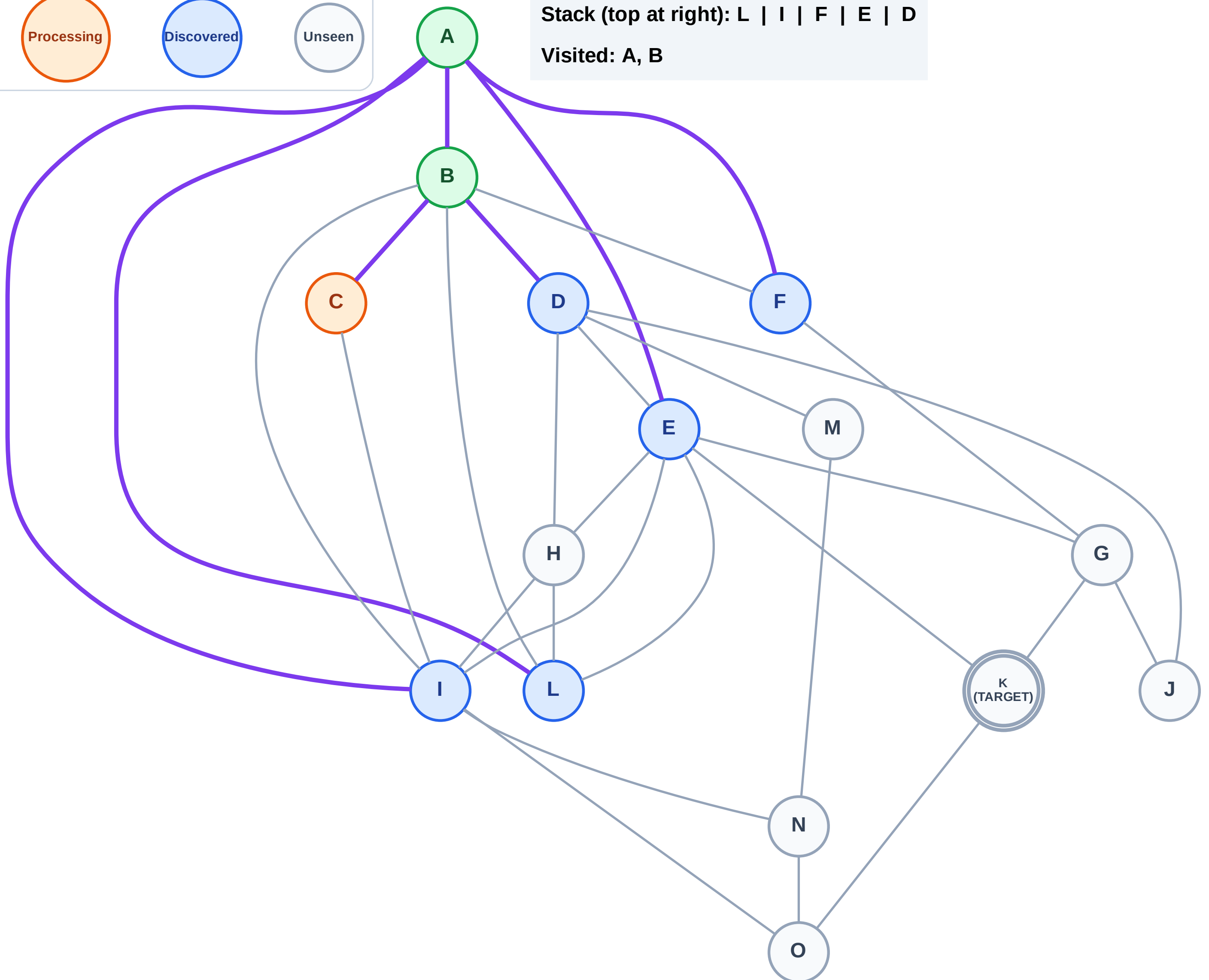
Discovered



Unseen

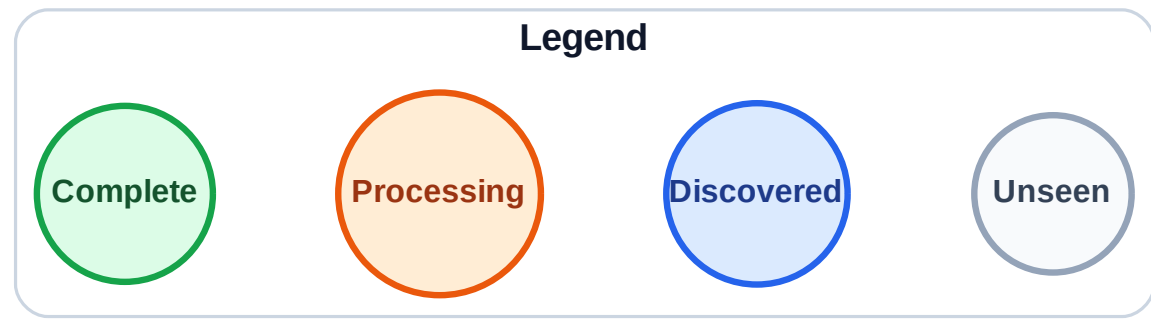
Stack (top at right): L | I | F | E | D

Visited: A, B



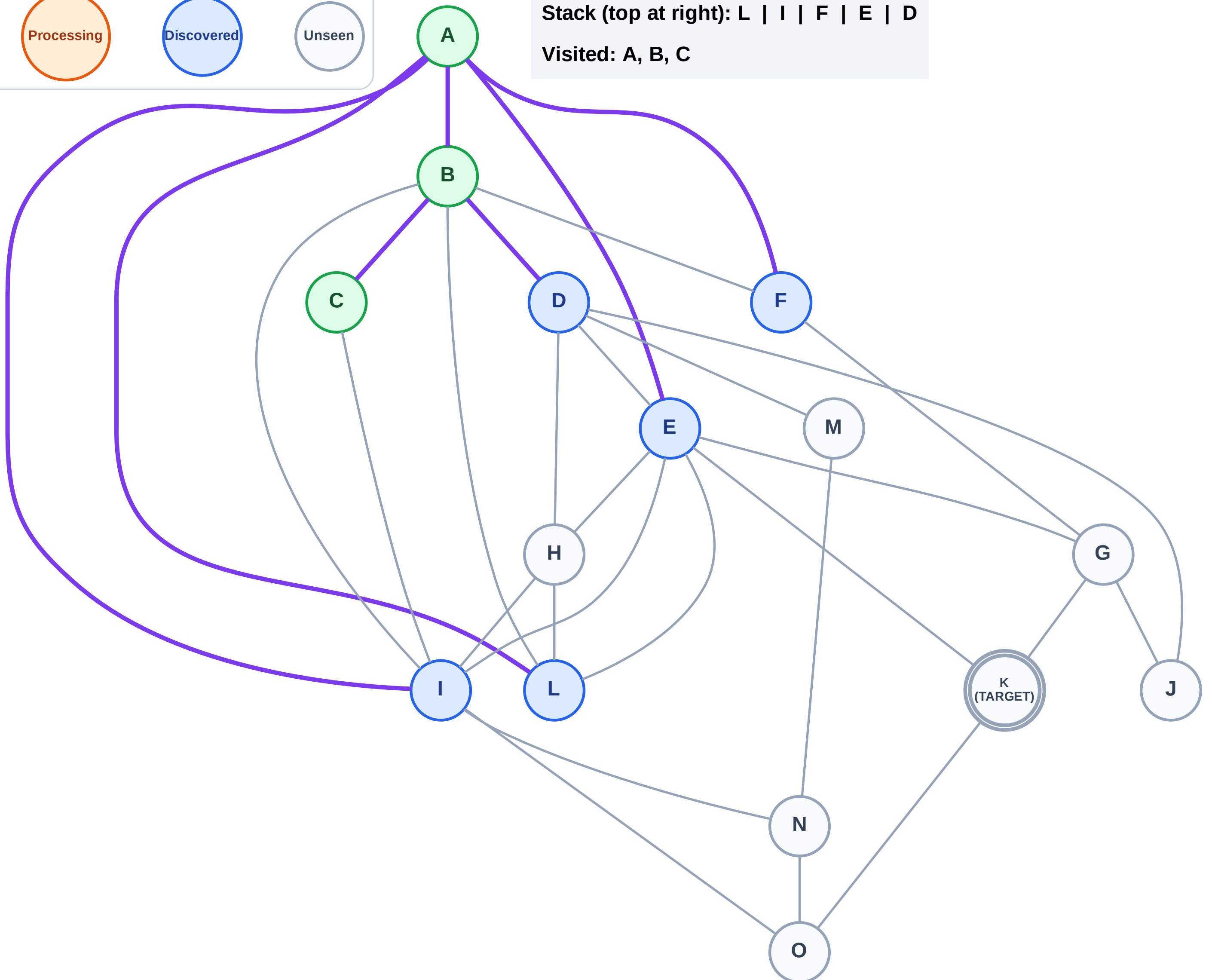
DFS Traversal

Step 7: No new neighbors from C



Stack (top at right): L | I | F | E | D

Visited: A, B, C



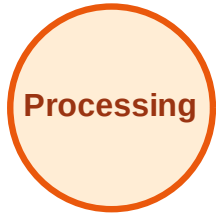
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



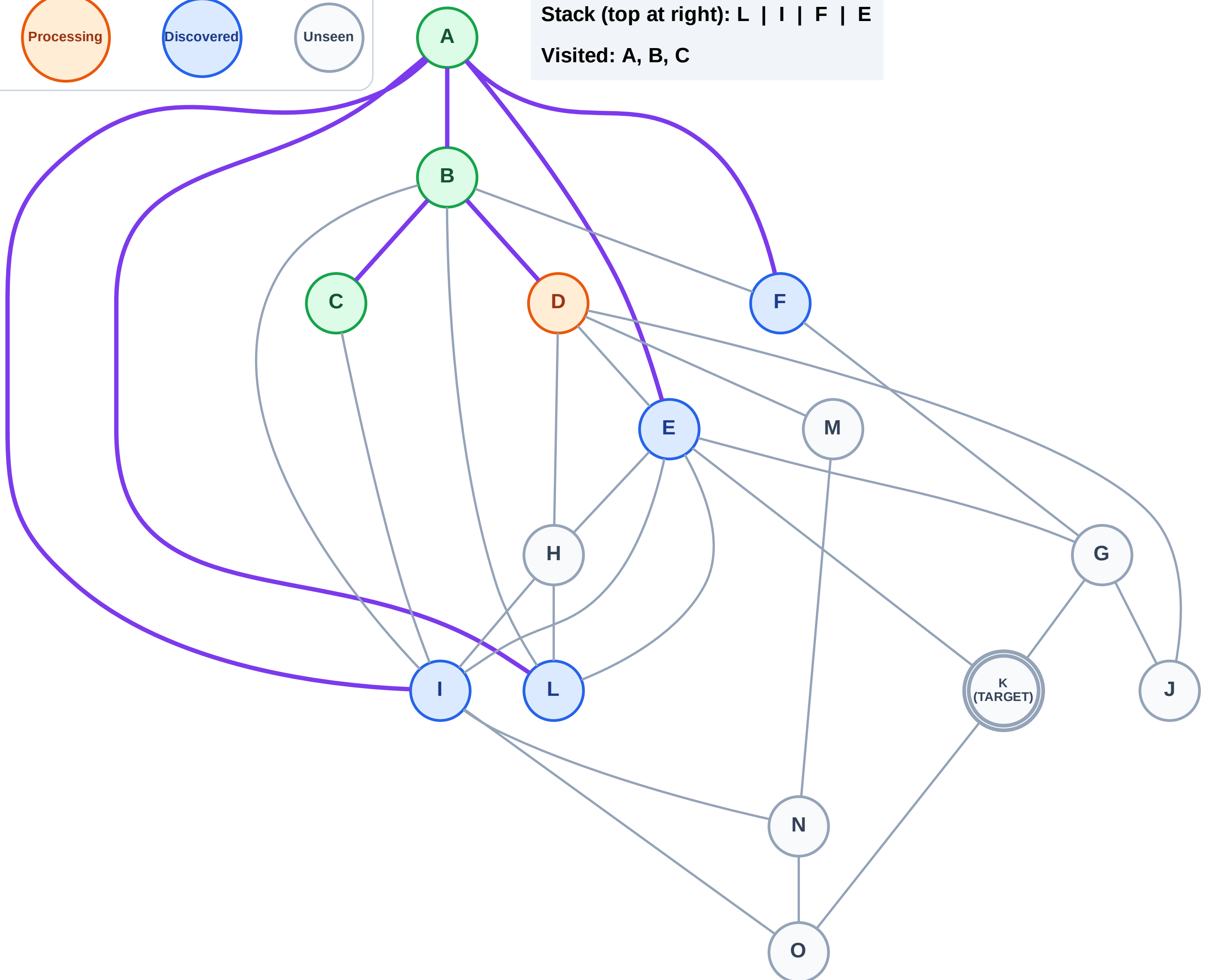
Discovered



Unseen

Stack (top at right): L | I | F | E

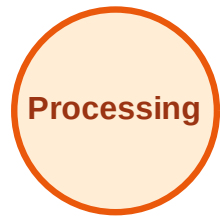
Visited: A, B, C



DFS Traversal

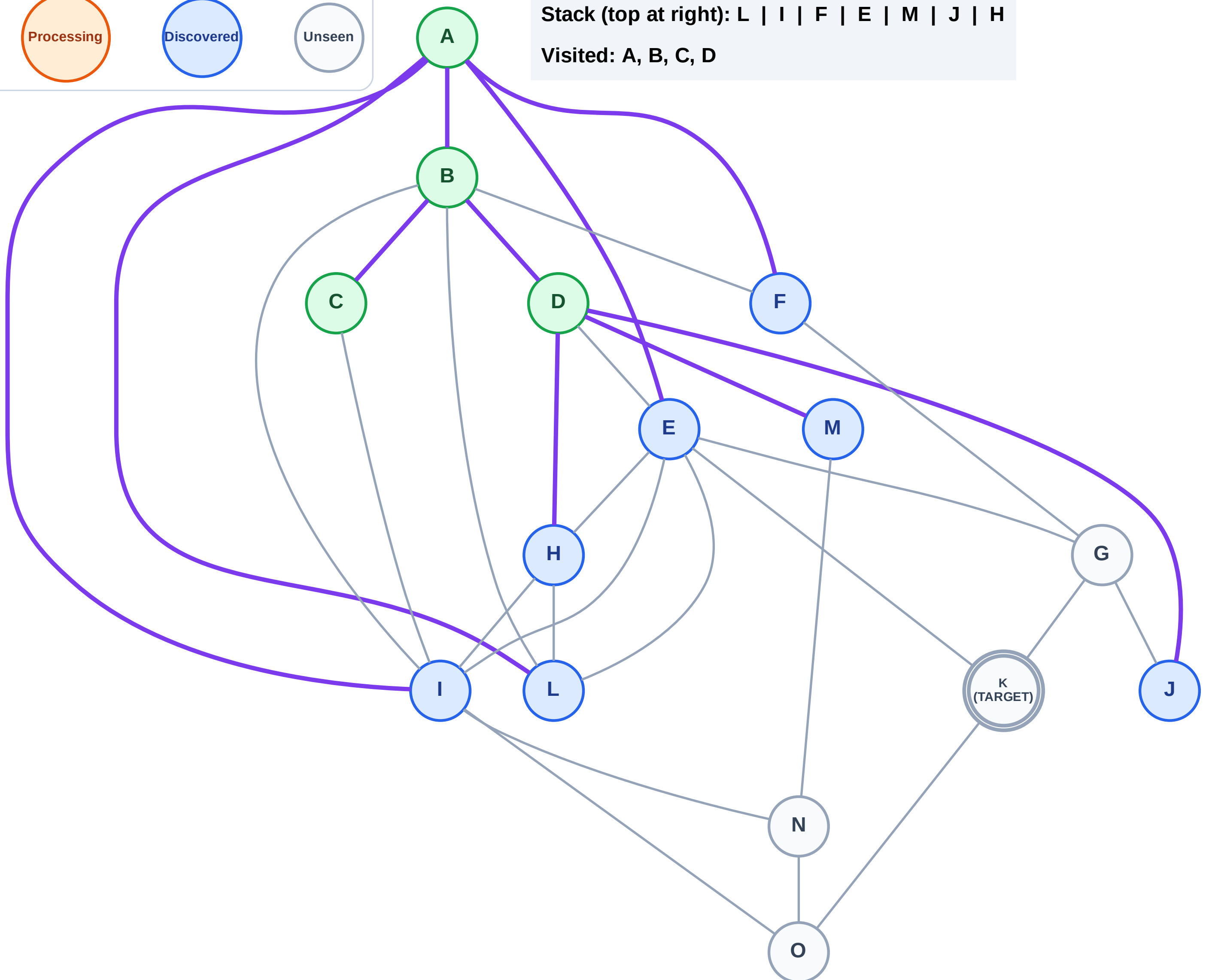
Step 9: Discover M, J, H from D

Legend



Stack (top at right): L | I | F | E | M | J | H

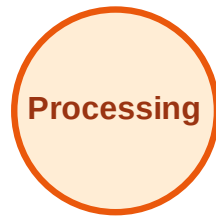
Visited: A, B, C, D



DFS Traversal

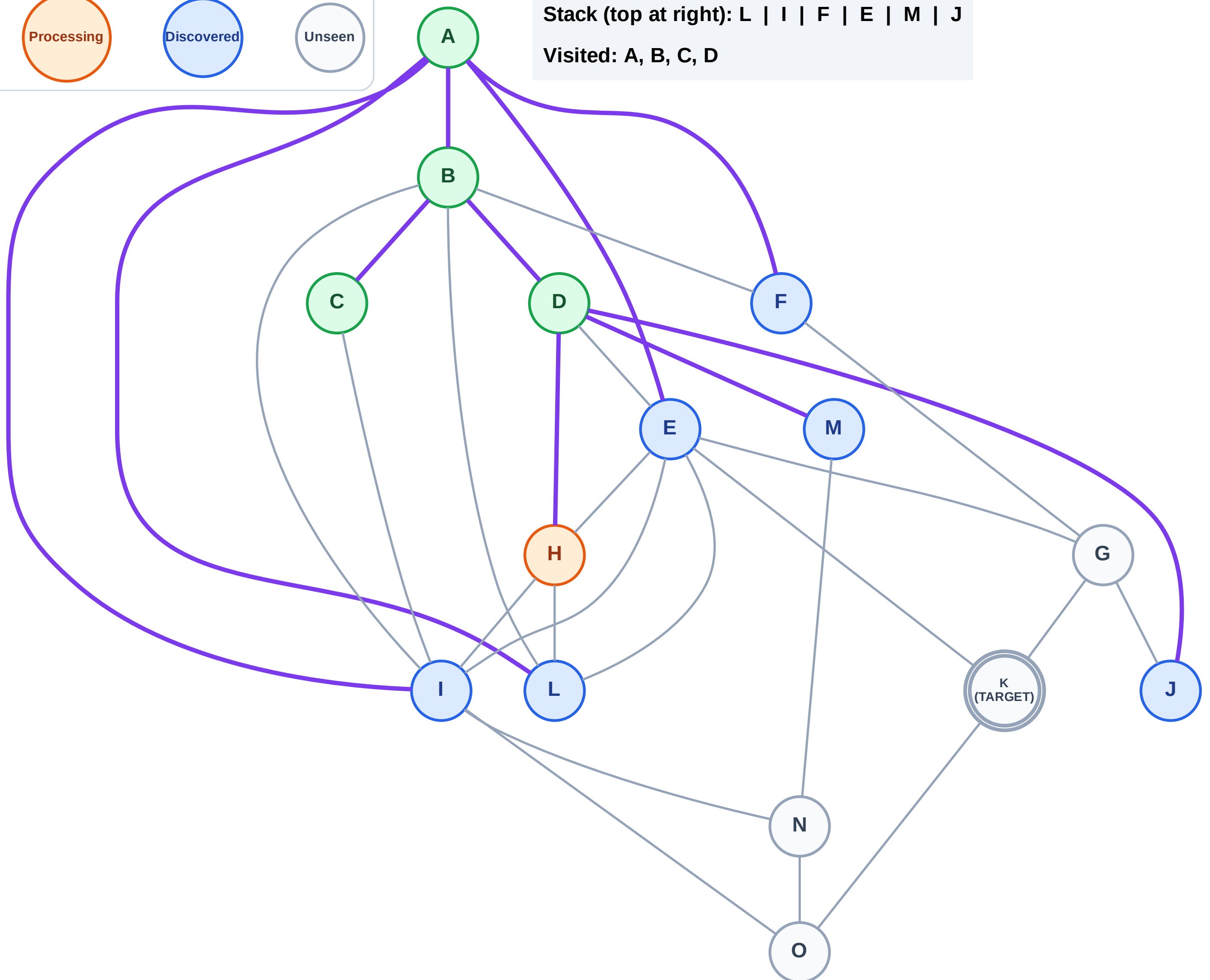
Step 10: Process node H

Legend



Stack (top at right): L | I | F | E | M | J

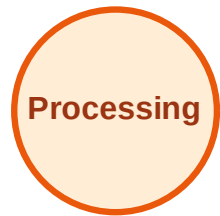
Visited: A, B, C, D



DFS Traversal

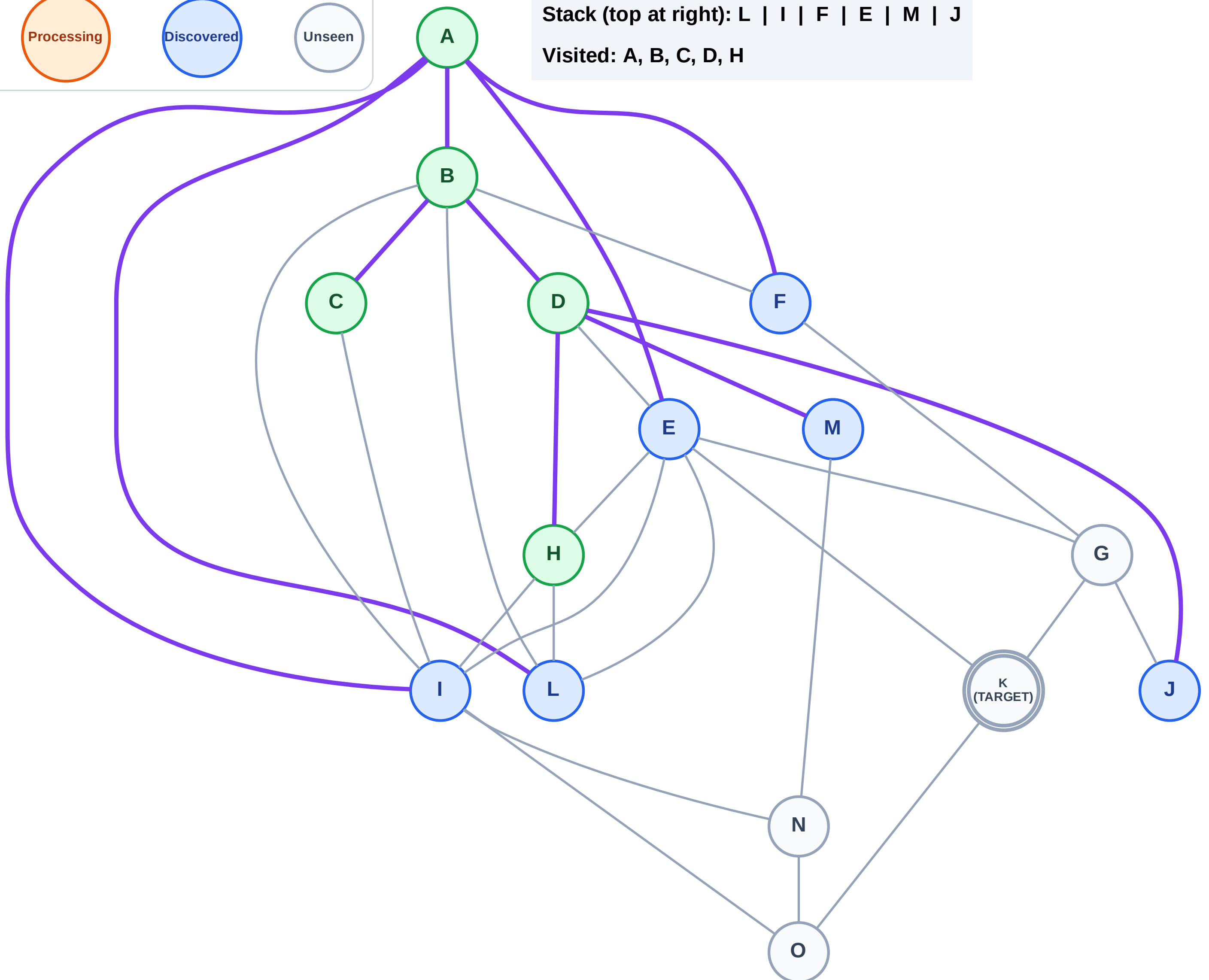
Step 11: No new neighbors from H

Legend



Stack (top at right): L | I | F | E | M | J

Visited: A, B, C, D, H



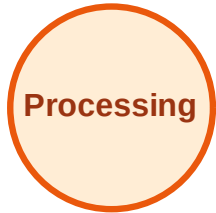
DFS Traversal

Step 12: Process node J

Legend



Complete



Processing



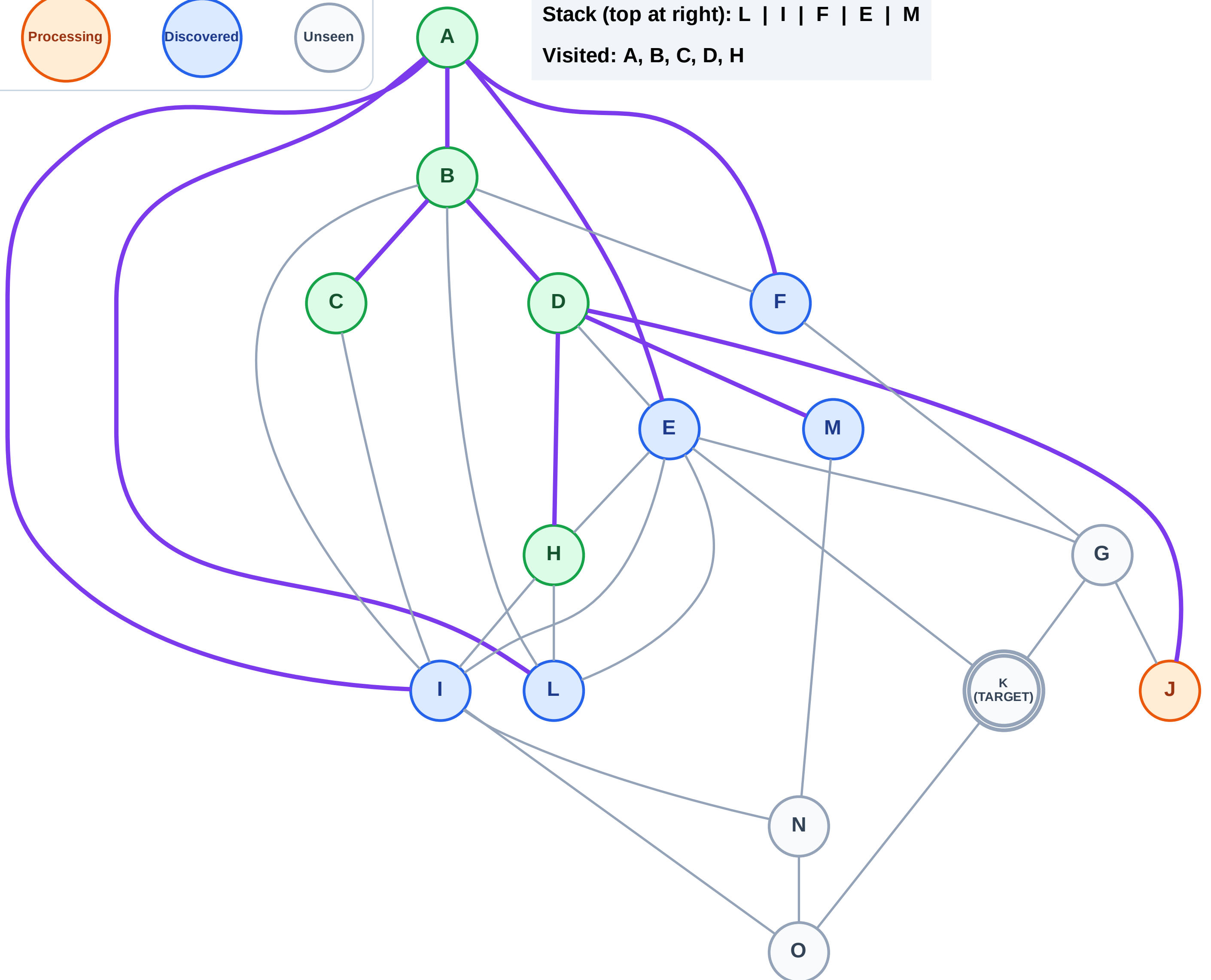
Discovered



Unseen

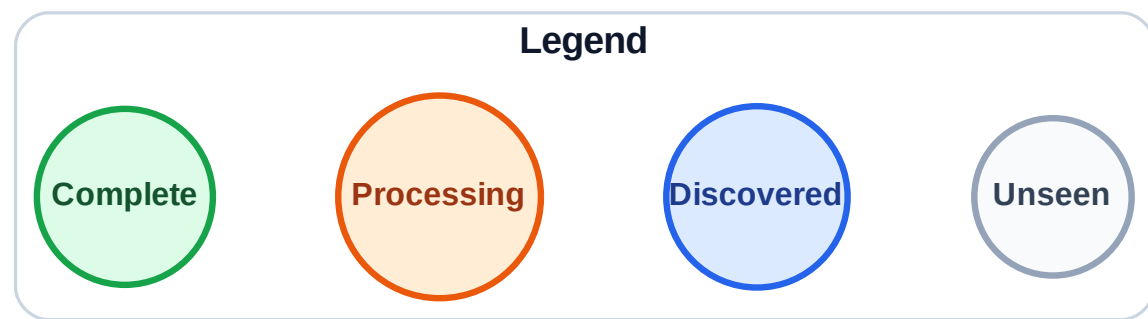
Stack (top at right): L | I | F | E | M

Visited: A, B, C, D, H

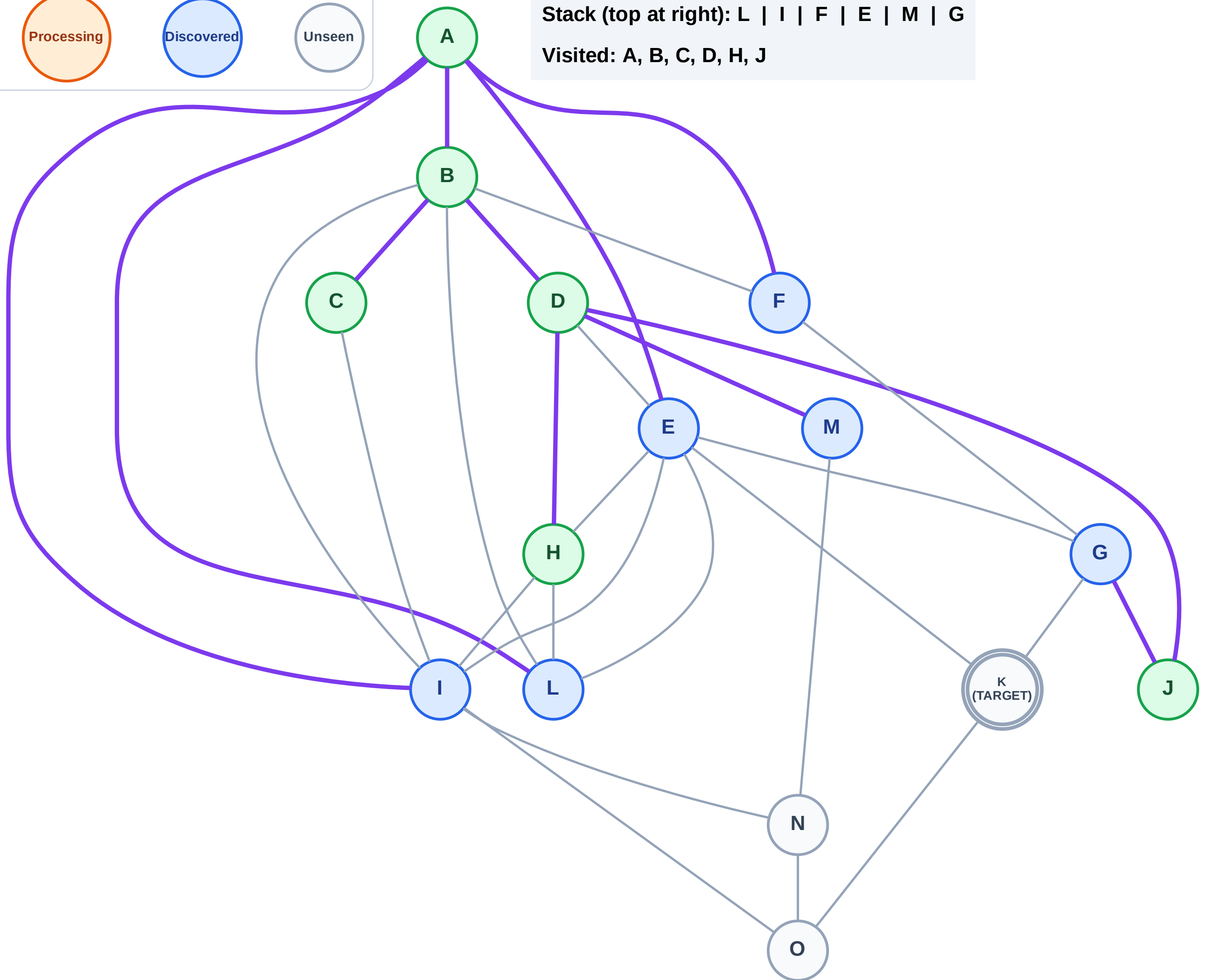


DFS Traversal

Step 13: Discover G from J

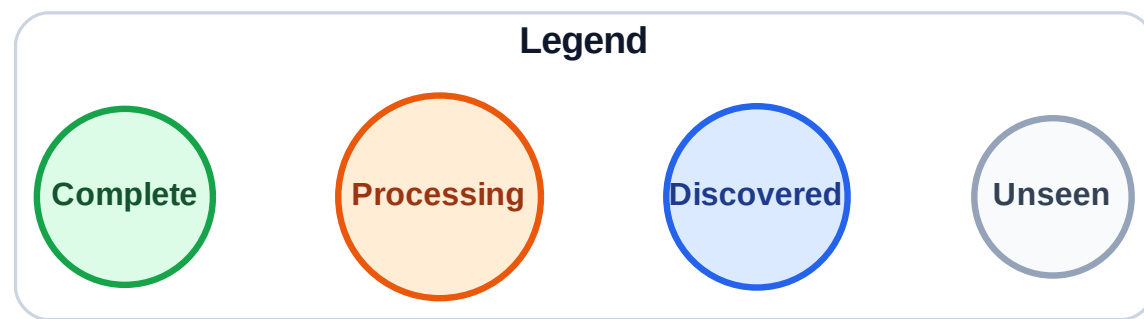


Stack (top at right): L | I | F | E | M | G
Visited: A, B, C, D, H, J

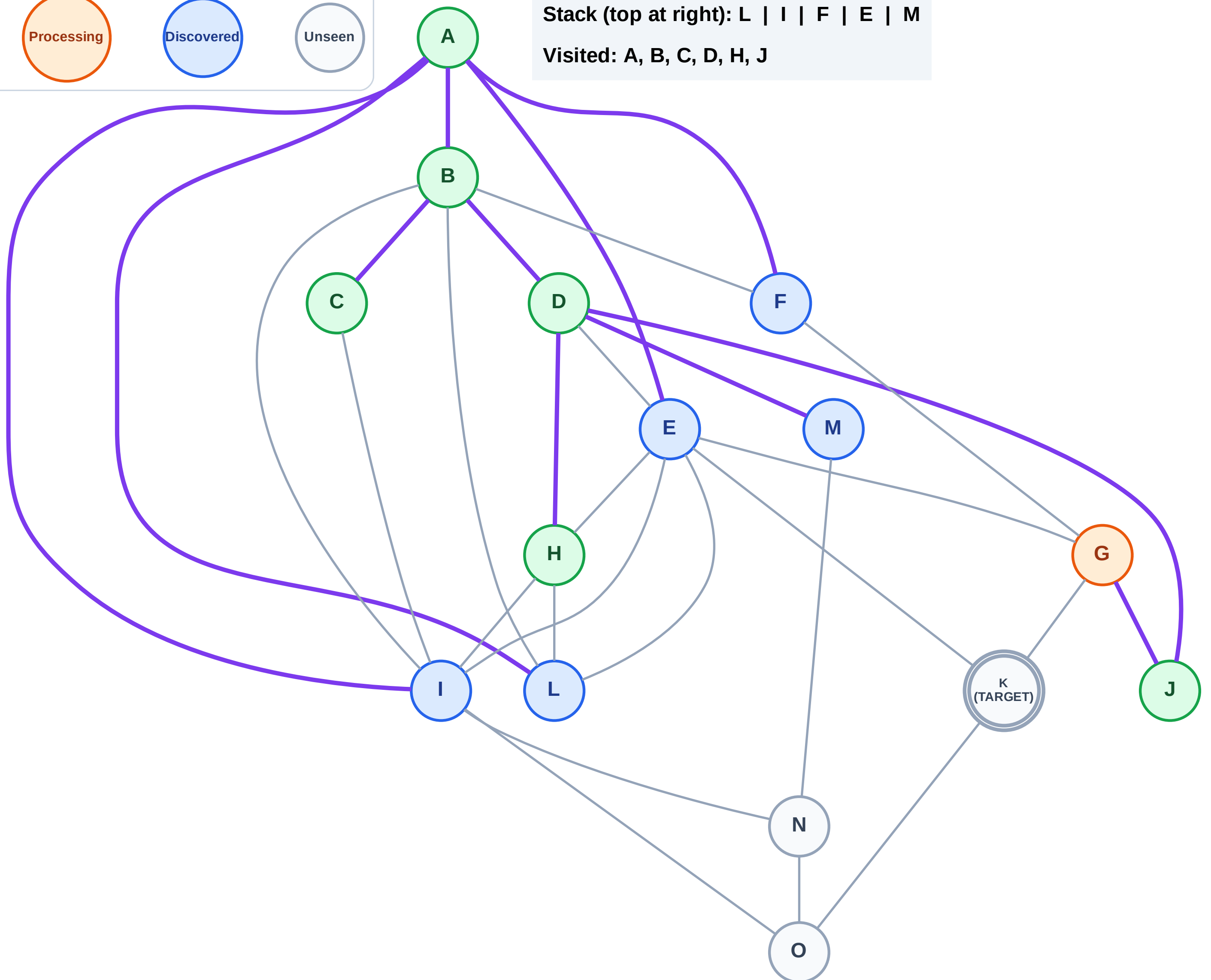


DFS Traversal

Step 14: Process node G



Stack (top at right): L | I | F | E | M
Visited: A, B, C, D, H, J



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (5 hops): A → B → D → J → G → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): L | I | F | E | M
Visited: A, B, C, D, G, H, J, K

